Chapter 11

Running replicable experiments

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Abstract
There is an increased focus on replicability in the social sciences. What does that mean for experimental design, analysis, and reporting in Economics? We cover topics most germane to the issue, from piloting, pre-registration, pre-analysis plans, power calculations, replication packages, and how to run a replication. For each topic, we define key terms, lay out what is currently expected or required, and describe what else can and should be done. We begin by discussing the state of replicability in Economics, highlighting that this is not an experimental-only problem; if anything, experimental work is the most replicable. We close with a discussion of what the profession could do to facilitate replication efforts.

1 Introduction

Assume your upcoming experimental paper will be replicated. This is good news. Your results are so impactful, assessing their veracity is all the more important. For science, and for your career, you want the replications to be done well. In this chapter, we attempt to prepare you for this eventuality. There will be a lot more detail, but really it all comes down to these two precepts:

Handbook of Experimental Methodology, Volume 1, ISSN 3051-1674
https://doi.org/10.1016/bs.hbem.2025.09.010

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1. Make sure it's right the first time.
2. Be very clear and very comprehensive in describing everything you did.

There are institutions intended to help you with this, like pre-analysis plans, and others that try to enforce this, like replication packages. There are numerous details you need to be clear and comprehensive about, like what did you pre-test or pilot, what outliers did you exclude, and so on. And there are other opportunities for you to describe everything, like in the paper itself or in comments in your code. We'll try to go through all of that here. We'll also walk through how to think about your statistical power even though this can be a challenge.

Before we begin, what does "replication" mean? There are three frequently used notions of the term. First, is "computational reproducibility", the idea that other researchers can achieve the same results with the same data and code. This is largely the goal of replication packages as required by journals today. Though many other fields who do not collect novel data are thinking of computational reproducibility when they say "replicability", we are more interested in actual replications—meaning testing the same hypotheses with new data. The researchers running these types of replications will need material such as instructions and software files (e.g. from z-tree, o-tree, or Qualtrics). While experimental economics has a long tradition of posting instructions in Appendices, the software is not always available, and the Methods part of the paper often does not contain all information needed to do an identical replication. For example, it cannot always be inferred in the paper how treatments were randomized within or between sessions or whether participants self-selected to sessions or not. Information related to running the experiment again can be thought of as "procedural replicability"—from your description and posted material, can we rerun the same experiment in a new sample? "Results replicability", on the other hand, refers to whether we can get the same results again when redoing the experiment in a new sample. Results replicability, since it is a measure of the credibility of the original results, is typically what experimenters mean when they say "replicability".

In this chapter, though we will mention replication packages and computational reproducibility, we will focus on ways to improve procedural and results replicability. In particular, we will discuss piloting, hypothesis pre-registration and pre-analysis plans, power calculations, and replication packages. We will also discuss ways to promote replicability, and thus credibility, for journals and the experimental economics community at large, including how to run a replication.

Our focus is on lab and online experiments, less so on field experiments. There has long been a discussion of whether enough replications are being published (see, e.g., Ankel-Peters et al., 2023; Berry et al., 2017) and both individual and large-scale replication projects in experimental economics and other experimental parts of the social sciences suggest that many results fail to replicate (see, e.g., Camerer et al., 2016, 2018). A number of solutions to replication problems have also been proposed. While there is not one silver bullet that can ensure replication success, there are many ways to promote replicability for experimental results, and procedural replicability should basically always be possible.

2 Experimental results are no less, and likely more, credible

We want to emphasize that there is no reason to think that experimental economics performs worse than other parts of economics when it comes to credibility; it is in fact likely to be the opposite. For example, Brodeur et al. (2016) find conclusive evidence of p-hacking in non-experimental empirical economics but inconclusive evidence in experiments. To identify this, they plot out the z-statistics from well-published, recent empirical work. They find a missing mass in the distribution precisely where, roughly speaking, p-values just above 0.05 would be found and an abnormally high mass where p-values just less than 0.05 would be found. It would seem researchers are, e.g. reporting only the specifications that are significant and hiding those that are not, i.e. "p-hacking". That they find p-hacking in non-experimental is perhaps surprising. This type of p-hacking is theoretically easier in experimental work, where perhaps the researcher can choose from several correlated outcomes; whereas, non-experimental data often offers the choice between completely unrelated outcomes and thus another type of p-hacking (see Simonsohn, 2020 for further discussion). Nonetheless, the evidence for this type of marginal p-hacking is more conclusive for non-experimental empirical economics. Brodeur et al. (2020) also compare different methods and find less p-hacking in papers using randomized control trials, "RCTs", and regression continuity designs compared to those using differences-in-differences and instrumental variables. Though we will discuss evidence that there is a replicability problem in experimental economics, all of the current research suggests that credibility problems are worse for non-experimental economics.

All fields are working to improve, and with different methods come different problems and different solutions. One of the luxuries of experiments is that they can be replicated—other researchers can typically fairly easily redo the experiments and see to what extent the results are similar. This does not always apply to many other types of empirical work in economics. In addition, the solutions to replication problems are typically more straightforward for experimental data and other types of prospective data collections than for other types of data. We will discuss pre-analysis plans, for which there are pros and cons. Pre-analysis plans are possible to credibly use in many parts of experimental economics, while they can be hard to use credibly where they are the most needed (e.g. for rich registry data, with many possible outcomes, many covariates, and so on).

3 Piloting

Pilots typically refer to any set of data collection not intended for publication. Pilots are often used for different purposes—testing to what extent participants understand the instructions, testing whether the software works properly, measuring means and variances for power calculations, testing whether there is an

indication of the hypothesized treatment effect, and so on.[1] Our advice is to utilize pilots to learn about confusion, software glitches, how long the study might take, and other related operational details only. For this chapter, we'll refer to this version of piloting as "implementation piloting"; whereas, any piloting where the researcher is looking at the data, e.g. to calibrate power, or to test the hypothesis, we will refer to as "data piloting".

Implementation piloting, i.e. to test how confusing your study is, whether the software works, and so on can be incredibly helpful. Too often, we can be overconfident in our ability to explain our study, forgetting perhaps that we might be using confusing econspeak or overly theoretically driven language. In such pilots, keep in mind the goal, and ask many open-ended questions: did you find anything confusing, what were you considering when you made this decision, were there any typos, did the software seem to work okay, and so on. We typically start with friends and students, but always make sure to do a small sample from the eventual study population (later excluding those subjects from the real study). Piloting beyond these ends is often done, but we do not recommend it.

Beyond this simple and practical use, pilots can be the most nefarious tool in the experimenter's toolbox. Perhaps the most often used, but unspoken, researcher degree of freedom in experimental economics is piloting different designs, different parameter calibrations, different language, different framing until you find something that works. Then you submit a pre-analysis plan for the new design, run the experiment, and publish a successful result. Without publishing the unsuccessfully used study designs, this gives an overly confident view of the robustness and generalizability of the result. This is bad for science. It might also be bad for the researcher, as replication work might show over their career that their results do not often hold up to different, but fair tests of the same hypothesis. Unfortunately, it can also be beneficial career-wise to pilot without reporting if this leads to more publications. And the urge to pilot might be stronger knowing you have to submit a pre-analysis plan: It might feel risky to submit a finalized data collection and analysis without having a better sense of the data. This may encourage "design-hacking" as termed by Frechette et al. (2022). In running many pilots, the researchers may tweak their design—the wording, the parameters, and so on—until they find a design that works. This may result in a true positive result, but whose robustness may be flimsier than the reader would expect. Though hard to quantify, it seems unreported piloting is currently quite common. We hope our field dis-incentivizes this practice in the future.

Related, though more obvious, every data collection needs to be decided ahead of time whether it is a pilot or it will be reported as an experiment in the

[1] We here focus on pilots for lab or online experiments, and not pilots for field experiments. For example, in development economics often some information sessions or qualitative interviewing might be necessary to get a basic understanding of how the population views the decision context, what norms or cultural aspects are important, and so on.

paper. Frequently, it is easy to convince yourself that an unsuccessful treatment didn't work for an arbitrary reason (e.g. subjects were particularly confused on this day), and it would be okay to move it to the appendix or not to mention it. To avoid exposing yourself to such very real motivated reasoning, we suggest the "planned experimental design" approach from Roth (1994) as opposed to the "independent trials" approach. As much as possible, try to plan out the entire study, e.g. a 2×3 factorial design, rather than piecemeal trying one arm, then another. The former commits you to a plan—and will also lead to proper randomization of participants to treatments which surprisingly is sometimes overlooked in experimental economics. The latter, as said, exposes you to motivated reasoning where it will be very tempting to ex post refer to some collection as a pilot, and involves multiple hypothesis testing that is rarely accounted for.

Finally, we suggest that it does not make much sense to pilot designs to attempt to measure potential treatment effects, or means and variances. Why? Pilots typically mean small samples. Small samples typically mean low statistical power. Low statistical power means noisy data. Pilots are typically run on small samples to get some idea on effect sizes and whether the design "works" in getting effects. Statistical power is a function of the expected effect size if the alternative hypothesis is true, the variance in the data and the number of observations. With low power, not only are we more likely to get false negative results, but statistically significant results are also more likely to be false positive ones (Ioannidis, 2005), and effect sizes are likely to be exaggerated or inflated or compared to, or even in the opposite direction of, the true effects (Ioannidis, 2008; Gelman and Carlin, 2014). Taken together, pilots do not make much sense as a means to understand potential treatment effects. They could potentially be useful to get some sense for appropriate parameter values and potentially to learn something about what the experimental costs might end up as.

Not everyone will agree with our take on pilots. In any case, we believe that increased transparency in piloting is clearly needed as in most cases from published papers it is not clear whether the authors had run a pilot or not, and if the former what the purpose of the pilot was. Pilots could be pre-registered (more on this below) as the timing of the decision that something is a pilot versus a real study is not always transparent, but if they are not they can still be transparently discussed to avoid e.g. "design-hacking" (see Holzmeister et al., 2024 for more on this concept).

4 Pre-registration and pre-analysis plans

In most other fields beyond economics, the concepts of pre-registration and pre-analysis plans roughly mean the same thing (see, e.g., Nosek et al., 2018). In economics, they can be used to mean the same thing but sometimes not, leading to some confusion.

Hypothesis registries are as they sound: An online database of hypotheses that have been investigated. The goal of such registries is to attenuate the "file-drawer problem". Too often when a researcher finds a null result, they do not publish it. In this way, the publication record is a biased sample, inhibiting our ability to make good inferences There is a literature proposing methods to measure and correct for publication bias; however, recent evidence comparing meta-analyses and large replication projects suggests some problems with these corrections (Kvarven et al., 2020). Hypothesis registration does not provide outcome data, but at least it helps correct this bias by letting us know how many times a hypothesis has been tested (if the registry is actually used). A hypothesis registration is sometimes accompanied by a pre-analysis plan.

Pre-analysis plans are an online, un-editable registered outline of how you plan to collect and analyze data. The goal of pre-analysis plans is to allow researchers (as both producers and consumers of research) to differentiate between confirmatory and exploratory tests, and to avoid the many "researcher degrees of freedom" in the analysis that can easily lead to statistically significant results that are likely to be false positive, which we describe in more detail below.

Most top journals require hypotheses from experimental papers to be registered. Though most would prefer the registration to be prior to data collection and analysis, they do not currently require it. To our knowledge, the *Journal of Political Economy: Microeconomics* is the only journal that currently requires pre-analysis plans for all experimental submissions (and encourages it when possible for other types of empirical studies), though it also allows for exceptions.

As pre-analysis plans are largely a response to the recent rise in skepticism of experimental results (though we believe that the skepticism should actually be larger for non-experimental results), it is worth noting where the skepticism came from, and the design and analysis elements that it is focused on. Starting in the field of psychology, with papers like John et al. (2012) and Simmons et al. (2011), they identified "researcher degrees of freedom" that were commonly used and assessed how dramatically they can increase the rate of false positives in published research. Common researcher degrees of freedom identified were, for example, researchers can stop collecting data once they have statistically significant results, can include many measures or treatments but only present those where there are statistically significant results, can exclude observations, can test interactions (e.g. do women respond to treatment differently?) or can use control variables to find statistically significant results. This is sometimes referred to as "p-hacking". In a related vein, Gelman and Loken (2013) discuss "forking" as a practice where researchers in "the garden of forking paths" choose to wander down one specific path in the analysis depending on what the data look like. This is to some extent a multiple testing problem where the number of tests is endogenous to the data—it can be forking with just one test since if the data would have looked differently more tests would have been done. It is also interesting to note that related concepts have been around in economics

for quite some time, such as Leamer's (1983) paper "Let's take the con out of econometrics", but did not have the same consequences on actual research practices as the Simmons et al. (2011) and Gelman and Loken (2013) papers. Later large replication projects, (e.g. Open Science Collaboration, 2015 and Camerer et al., 2016, 2018) showed replication rates much lower than published p-values would suggest assuming reasonable power, consistent with the purported use of researcher degrees of freedom or forking. As a result, many have proposed the use of pre-analysis plans to pre-commit researchers to a design, collection, and analysis—i.e. avoid both p-hacking and forking—defending them from subconscious as well as conscious motivations to deviate.

In 2012, the American Economic Association (AEA) launched a randomized control trial, "RCT", Registry. While there is room for detailed information about the experimental design, hypotheses and statistical tests, it is also possible to not say much when registering a study here. It is also possible to register studies after the results are known. Thus, when it says in a paper that a study is registered at in the AEA RCT Registry it can mean many different things.

When a pre-analysis plan is posted, on the other hand, it by definition needs to be posted before any analysis has been performed, and ideally before any data has been collected. A pre-analysis plan should contain information on design, inclusion criteria of participants, variables and analyses. This information should be detailed in the sense that design and analytical choices should be made in the pre-analysis plan to avoid "researcher degrees of freedom", thus the analysis details should include information such as how variables will be coded, how the exact analysis will look like (regression models etc.) including whether potential control variables will be coded and used and whether standard errors will be clustered in any way. The pre-analysis plan does not need to contain any literature review or justification for why the researcher has chosen some specific design or analysis. When there are several tests, we also believe that it can make sense to set up a hierarchy, where the most important ones are labeled primarily hypothesis tests, followed by secondary hypothesis tests, robustness tests, and pre-registered exploratory tests. Pre-analysis plans can also cover to what extent there have been pilots (that could be covered by the same pre-analysis plan or separate ones that are linked to).

There exists a number of templates for pre-analysis plans, and many outlets where you can submit pre-analysis plans. We include those references at the end of the chapter. Generally, you will be asked for a short description of your hypothesis, when and where you're collecting data, what your outcomes are and precisely how you are measuring them (e.g. with an index, or a normalized value, and so on), what your design is generally and how many treatments you are running, the level at which you are randomizing and how, who your sample is, the number of observations (with room to justify, but not necessary), and do you have IRB approval or exemption.

There are pros and cons of using pre-analysis plans. As discussed above, one pro is to eliminate your degrees of freedom in the analysis to get credible results

if you are testing hypotheses. Pre-analysis plans should thus allow the reader to differentiate between confirmatory and exploratory analyses. Both types of research are important. Indeed many of the most important discoveries in the history of science were more incidental than intentional. Nonetheless, it is important to delineate the two for the reader—as well as for the original researcher. As it is possible to pre-register exploratory tests, the final paper can also differentiate between pre-registered vs non-pre-registered exploratory tests. The use of pre-analysis plans should of course not prohibit the analysis and reporting of exploratory tests—these can indeed be hypothesis-generating and lead to novel discoveries. But every researcher should keep in mind that many reviewers and editors (including Anna) consider the evidence value of non-pre-registered tests to be weaker at the individual paper level.

There are (at least) two paradigms of discovery. One is that discovery should happen at the paper level. This is arguably very true for large field experiments. Another paradigm is that discovery occurs at the literature level: Original findings always need to be verified later, both in terms of truth in direct replications, and in terms of robustness. We see the value in both. We hope as researchers you aim for as much rigor and robustness as possible in one paper, but also allow for discovery to be a longer process when necessary. Pre-analysis plans may also be helpful for informing Bayesian readers about your results, and also so that you yourself does not fool yourself into thinking that what turned out to be exploratory instead was confirmatory.

Currently, however, in addition to there being confusion on what pre-registration means in economics, there is also no clear norm in economics on how to report whether the pre-analysis plan is followed or not (see Ofosu and Posner, 2023 and Abrams et al., 2020 for evidence that pre-analysis plans and pre-registrations are not always strictly followed and deviations are not always delineated). Even with the lack of strong norms of what should be included in a PAP, the evidence suggests they at least partly work, and do so beyond a pre-registration alone. Brodeur et al. (2024) explore to what extent there is evidence of p-hacking and publication bias in terms of reported test statistics in RCTs published in 15 leading economics journals 2018–2021, comparing papers that are not pre-registered, are pre-registered without pre-analysis plans, or are pre-registered with pre-analysis plans. The results suggest that pre-registration in itself does not help curb p-hacking, while pre-registered studies that have complete pre-analysis plans are less p-hacked. With more details in the pre-analysis plan and greater transparency in how it is followed or deviated from, we believe that p-hacking or forking could be curbed even further.

If you are going to submit a pre-analysis plan, we suggest you make it meaningful. At the risk of sounding repetitive: Be thorough and comprehensive. Be as precise as you can about your experimental design, your plan for data collection, and how you're going to analyze the data. This does not necessarily mean choosing, e.g., a regression and set of controls ahead of time if for some reason this is not feasible. Much of the pre-analysis plan can be dynamic, as long as it

ties your hands to avoid researchers degrees of freedom. As suggested by Olken (2015), you can lay out how you plan to choose covariates only once you have the data, or the process by which you might exclude outliers or confused subjects. Lay out what you can, even if it's just a process rather than a determined plan, and try to avoid running into any ambiguities at a later stage. And if it sounds difficult to envision what the analysis can look like, you can potentially use synthetically generated data for inspiration. Finally, in the paper, be clear about what is in the pre-analysis plan and what is not. Describe why you went beyond the pre-analysis plan if that is what you do. And we believe your reader will appreciate knowing what was ex ante versus ex post.

Important for understanding how your referees and editors might view your work, experimental economists are largely, but not wholly, in favor of pre-analysis plans. In 2023, Imai et al. (2024) ran a survey targeting the Economic Science Association (ESA). Their sample includes 519 individuals at all career stages. 66% of respondents report being either somewhat favorable or very favorable to pre-analysis plans, while 23% report being somewhat or very unfavorable. More junior researchers, as measured by year of PhD, report being more favorable to pre-registration. In contrast, the support for replications is almost unanimous—93% of respondents report being either somewhat favorable or very favorable to replications, while about 2% report being somewhat or very unfavorable. There is also some discussion of whether replications are more effective than pre-analysis plans at removing false positives from the publication record (Coffman and Niederle, 2015). A simple calibration of the model in Coffman and Niederle (2015) (which is a modest extension of Ioannidis, 2005) suggests when replications are feasible, they might be more effective than pre-analysis plans, and the added value of pre-analysis plans when replications are done is limited. However, this is one calibration of one simple model, and replications are not yet commonplace even for the most impactful work (Berry et al., 2017). And if referees appropriately appreciate exploratory analysis, the downside of pre-analysis plans likewise seems limited. Somewhat related, it could also be argued that pre-analysis plans are not as needed when experiments are theory-driven and the theory clearly tells the researcher how to analyze the data.

Finally, there are other advantages to utilizing pre-analysis plans (this section is largely inspired by Olken, 2015). First, we find it very useful to think through all aspects of the design and analysis ahead of time (and have in one instance written the analysis code at the time of the pre-analysis plan which for that project likely saved time). Without fail, thinking through the tests we are going to run, changes the design, or adds some additional questions or measures. Second, a pre-analysis plan can give you latitude to do analysis that would otherwise be considered suspect. Know you're going to have extreme outliers or a lot of confused subjects? Just say you'll remove them (or whatever correction you prefer) in the pre-analysis plan, and you're good to go. Third, a pre-analysis plan can help solve any ex post agreements. Maybe two co-authors may have

had differing opinions on what the data show, or maybe your implementing partner organization really wants to tell the press you found a treatment effect. You can help these situations by getting buy-in upfront on the fairest way to analyze the data.

Perhaps the most challenging problem related to pre-analysis plans is that they are so difficult to use where they are really needed. It is challenging, if not impossible, to write a credible pre-analysis for registry data, or data that is already available online, or data that you for some reason "have to get to know" before you know how to analyze it—i.e. for most types of non-experimental empirical economics. As discussions of pre-analysis plans for experimental papers become more common, we believe that it is of utmost importance to simultaneously discuss how to evaluate the credibility of non-experimental papers and suggest ideas of how to improve that credibility. If anything, this discrepancy in credibility between experimental and non-experimental papers should in our view make journals more, and not less, positive to experimental papers.

5 Power calculations and priors

Statistical power is the probability of finding a statistically significant result from your study given sample sizes and a set of assumptions about means and variances. For a power calculation, you can input N-1 parameters (e.g. power wanted, means and variances) and calculate the remaining degree of freedom (e.g. necessary sample size). Power calculations can be equally helpful and challenging. In this section, we will talk about the importance, challenges, and high-level to-dos, but unfortunately, we do not have the real estate to provide a more detailed technical guide.

Historically, power calculations have largely been absent in economics papers with lab experiments, but it is increasingly common. Power calculations are more guesswork, and perhaps noisier, than you might hope (e.g. Pek et al., 2024). But they can still be helpful. Let's walk through each parameter in a power calculation. Most tests are going to require two means, e.g. the average of the outcome in your control and treatment groups. As we said above, attempting to measure these in a small pilot can be unhelpful, so what can you do? You can look to the literature for similar effects (though see note below on potentially inflated published effect sizes). You can also simply plug in the smallest difference in means that you would find meaningful enough to headline your paper. For example, do you want to stand in a seminar room pointing at an effect size that is below 10%? 5%? This will be your "minimum detectable effect". The next parameter in your power calculation is sample size. Often, this is the desired output of the calculation (i.e. for a given desired power level and set of assumptions, how much data do I need?). Often though, there is a very real budget constraint that drives this parameter: I can afford *this* much data. That is fine—this can be how you justify your sample size but even then it can make sense to do a power analysis so you can have reasonable expectations of what

you can achieve (and potentially cut down on the number of conditions to get better power). The trickiest parameter is often the variances. If your outcome is not binary (hence with a variance pinned down by the mean), what can you do? Again, you can look to the literature for any measures related to your own. You can experiment on power calculations using a wide range of reasonable variances to see how palatable the power or sample sizes are. You can also think about effect sizes in standardized terms such as Cohen's d, and consider what would be the minimum Cohen's d you would consider meaningful. Finally, if your data collection is dynamic, you could consider pre-registering stopping data collection when you are powered to find a specific minimum detectable effect. That is, rather than pre-specifying a given sample size, N, to collect, you can pre-specify that you will stop when the standard errors are such that you would expect to find an effect at the pre-specified significance level with a probability equal to the pre-specified power. If so, we recommend doing this with blinded data, i.e. randomly assigning treatment labels but keeping the data the same otherwise, an approach promoted by Olken (2015).

Power is not only useful to know if your study is a good use of time and money, but it is also important for replicability. Published positive results from a low-powered study are more likely both to be false positives or to be an exaggeration of the true effect size. This is simply a result of having large variance in outcomes from the small sample and then taking the conditional expectation of the effect size being positive and statistically significant. So your results might not replicate, and if they do, the effect sizes will be smaller. This is also important to consider when you're evaluating effect sizes in the literature. Exaggerated effect sizes create problems for ex-ante power analyses when you are planning your experiment. If you base the power calculation on a previously published somewhat related result, or even a meta-analytic effect size, you might be underpowered and fail to detect a true positive effect. The results from high-powered replication projects (Camerer et al., 2018; Holzmeister et al., 2025) suggest that replication effect sizes of results that successfully replicate, in the sense that the replication effect is in the same direction as the original result and it is statistically significant at $p < 0.05$ in a two-sided test, is about 75% of the original effect size. Taking this into account, having 90% power to detect somewhere between $1/2$ to $2/3$ of an original effect size can be a good heuristic if the budget allows it. This leads to sample sizes that typically are substantially larger than the original ones, but if the effect size used in the power analysis is very large and highly statistically significant, it can lead to smaller sample sizes.

Whether results will replicate will also depend on the priors of the tested hypotheses—what probability do these hypotheses have of being true in the first place? One way to assess these priors can be to do forecasting surveys where researchers are asked to assign probabilities of a result replicating or being statistically significant (e.g. Dreber et al., 2015; DellaVigna and Pope, 2018; DellaVigna et al., 2019; Landy et al., 2020). Such surveys have become

increasingly common in the social sciences in general, and are sometimes used to combat publication bias (as in helping the researchers show that for example a null result from a RCT was something the community of researchers did not expect) and sometimes used to say that replication results can be predicted (as in there being something systematic about results that successfully vs unsuccessfully replicate and this is something community of researchers can predict). In either case, it makes sense to consider the timing of the survey and perform the survey before the data is collected so that other researchers do not start thinking that if a survey is sent out it is to show that the result was an unexpected null result.

6 Replication packages: computational reproducibility

When you publish your paper, you are almost surely going to have to submit a "replication package". As of July 2024, every one of the top general interest journals we audited—*The Quarterly Journal of Economics, The American Economic Review, Econometrica, The Journal of Political Economy, The Review of Economic Studies, The American Economic Journal: Microeconomics, The American Economic Journal: Economic Policy, The American Economic Journal: Macroeconomics, The Journal of Political Economy: Microeconomics*, and *The Journal of Political Economy: Macroeconomics*—all required replication packages to be submitted upon acceptance of a paper. Replication packages should be a no-brainer to experimentalists—we rarely have any confidential data that cannot be shared, and the code typically does not require specific and complicated computing environments.

A "replication package" is a bundle of materials sufficient to allow a third-party researcher to replicate the experiments and reproduce the analysis. Some journals' requirements are simply to provide a data set and an analysis file. We suggest you go beyond this. The data you provide should be the raw data, e.g. the data files you downloaded from, e.g. Qualtrics before you merged or cleaned anything. Your analysis files then should both clean and assemble your data in addition to performing the empirical analysis reported in the paper. Explain, using comments, everything you do, from dropping duplicates, dropping those who did not finish, coding this variable this way, and so on. The idea of preparing a replication package for your work is useful to consider early on in a project—many of us have realized that our code from a few years ago is not always that understandable even for us who wrote it, but writing it with your future self or for others in the first place should improve this. As Lars Vilhuber (AEA Data Editor) often emphasizes—having "computational empathy" is a useful thing.

The posting of raw data and analysis code in replication packages allows other researchers to test to what extent they can computationally reproduce the results. This is a lower bar to pass than replicability, yet there are studies suggesting not only has this been a problem in the past in economics (e.g., Dewald et al., 1986) but it has also been a problem more recently (e.g., Gertler et al., 2018; Herbert et al., 2024). This will however be less of a problem in the future as many journals require that the posted analysis code when run on the posted data delivers the results in the paper. Checking this is the role of Data Editors that are becoming increasingly common at economics journals.

7 Running a replication

Next time you read an experimental paper, consider whether you, based on the information in the paper, would be able to set up a replication. Are the instructions, software files (from e.g. z-tree, o-tree, Qualtrics) and information about the setup of sessions and conditions provided directly in the paper or through links from the journal or authors? Is the experimental procedure clear—were participants randomized to conditions between or within sessions, and if the former, was randomization really at the individual level or at the session level? Over how long a time period was the experiment run? This type of information and material can ensure procedural replicability. Consider this for others who might want to replicate your results, and at some point you might want to do a replication study yourself. Many researchers also share this information if you ask them about it for a replication, but the more the original paper mentions all of this the better.

If you do a replication, or if someone else runs a replication based on your research, this could be done in a direct replication (sometimes referred to as an "intended replication") or in a conceptual replication (sometimes referred to as a "quasi-replication" or "robust replication"). In a direct replication, the idea is to have the replication as close as possible to the original study in terms of the design and analysis. In a conceptual replication, the idea is to test the same hypothesis as in the original study but some thing(s) can be altered in order to see whether the results are similar or not, e.g. in the intervention/treatment, measures, or analysis. The line between the two types can be a bit blurry, and if you are to conduct a conceptual replication of someone else's work it can make sense to discuss your design and choices with the original authors before any data is collected, if possible. Many studies test several hypotheses, and some papers contain several studies. We suggest that at least initially it makes sense to focus on one study and one central result (and base the power calculation on this result, while other results can of course also be studied), and plan how to replicate this in either a direct or conceptual replication. The *Institute for Replication* (I4R) has a replication report template for such a replication (https://osf.io/fc2ga) that builds on the replication work from Camerer et al. (2016, 2018) and Holzmeister et al. (2025). This replication report template covers aspects such as the original study and hypothesis being replicated, the test statistic of this hypothesis, the criteria for replication (typically what is below referred to as the statistical significance indicator), the sample (including inclusion and exclusion criteria), the procedure of the replication, as well as the planned analysis. There is of course a section on statistical power, and power calculations tend to be more straightforward when you do a replication as you have an effect size to build on. We suggest that you standardize the original effect size and aim for 90% power to detect 1/2 or 2/3 of the original effect size at $p < 0.05$, and if this results in a sample size smaller than the original one we suggest that you use the original sample size. In many ways, this replication report resembles a pre-analysis plan.

What is then the difference between a conceptual replication and an extension? These are close neighbors. One way to think about it is the following: Let's say you are largely varying one parameter from the original study. If a careful, fair reading of the original manuscript suggests the result is robust to that parameter, you are running a "conceptual replication". If the original manuscript suggests this might change the result, you are running an "extension". For example, Henrich (2000) finds the Machiguenga people make ultimatum game offers ~15% of the total pie. Henrich claims this is due to local culture or norms. Typical lab studies in the US instead find ultimatum game offers of 40–50% of the pie. These papers typically do not mention culture or norms. If you were to run an ultimatum game study in, say, the Dominican Republic, that would be a conceptual replication of the US studies, but an extension of Henrich.

Several indicators of whether a result replicates have been proposed (see Dreber and Johannesson, 2025 for a more extensive discussion). For experimental replications, the most commonly used one is the statistical significance indicator, which is a binary indicator of whether the replication result is in the same direction as the original result and statistically significant, or not (and most original results are communicated also in this binary way). Another common indicator is the relative effect size indicator, which is the replication's effect size divided by the original effect size. This indicator allows us to see to what extent original effect sizes are inflated. Both indicators can also be used for reproducibility studies (using the same data as the original studies), and can typically also be applied to studies with several tests. Other replication indicators have been proposed, including the "prediction interval approach" (Patil et al., 2016) that in a z-test tests for a statistically significant difference between the replication effect size and the original effect size, the "small telescopes" indicator (Simonsohn, 2015) that tests if the replication effect size is significantly smaller than a "small effect" defined as the effect size the original study had 33% power to find, as well as the Bayes factor of the likelihood of the original hypothesis vs the null hypothesis based on the replication data (Wagenmakers et al., 2018). These replication indicators also come with their pros and cons—for example the prediction interval approach is the most "generous" to the original studies that are the most likely to be false positives, and the small telescopes indicator assumes that all original studies are powered similarly, leading what is considered a small effect size to be larger for underpowered original studies. For original null results, Dreber and Johannesson (2025) recommend an adjusted version of the statistical significance indicator which estimates the fraction of statistically significant replication tests irrespective of the direction of the effect.

8 How can our profession otherwise improve replicability?

Much of this chapter is a how-to for the individual to improve the replicability of their own work. Let's also think about what the field of economics is doing, or can do to further promote replicable work.

8.1 Promoting, citing, and organizing replications

Chances are high that someone else will attempt to replicate one of your studies at some point. I4R has the goal to try to systematically reproduce and/or replicate as many studies as possible from recent publications in leading journals in the social sciences, with a particular focus on economics. I4R together with local hosts organize Replication Games, which are one-day events intended to encourage reproducibility studies and replications and make them more fun (Brodeur et al., 2023). Journals are also increasingly positive to replication, with some explicitly mentioning this (e.g., List, 2023). There are also proposals on how to promote replications by journals (see Drazen et al., 2021 for a proposal and proof of concept) or from different teams of researchers joining forces to replicate the results of one team and then becoming coauthors on a joint paper (Butera et al., 2020). Coffman et al. (2017), among others, suggest a norm of citing replication work alongside the original. Doing so could incentivize replications with the coin of the realm: citations. The promise of replication is thus stronger than ever, and an email from the I4R or any other replicator should hopefully not make anyone stressed but rather happy that others are interested in your study and results.

Related to the promotion of replications, researchers need to learn about replication outcomes. Replications can only be helpful to science, and cited by others, if future researchers can readily find replication attempts of original work. A couple of recent papers suggest that studies that unsuccessfully replicate are either cited more than studies that successfully replicate (Serra-Garcia and Gneezy, 2021) or are cited equally (Schafmeister, 2021). Coffman et al. (2017), among others, have suggested a replication database, where replication work, and their findings can be posted and easily linked to the original study. Through this repository, one could search any paper whose work they are building upon to see what the larger literature adds to the credibility of the result.

8.2 Registered reports

The idea with pre-analysis plans is to make research results more credible and more likely to replicate. However, it is not clear that results that follow pre-analysis plans are as likely to be published as results from experiments that do not have pre-analysis plans, everything else kept equal. Why? The former are more likely to result in null results. Thus pre-analysis plans are not a clear solution to the problem of publication bias. Registered Reports (Chambers, 2013; Nosek and Lakens, 2014), a relatively new publication format, can however help with this. Registered Reports resemble normal papers in many ways—but there are no results. When they are submitted to a journal, they outline the contribution (with related literature), the planned experimental design, measures, hypotheses, planned analysis etc—similarly to a pre-analysis plan extended with an introduction and literature review. This is what is peer-reviewed, so that reviewers can give feedback on all these parts without knowing any results, as

the data have not yet been collected. If the Registered Report eventually gets accepted, the journal commits to publishing the paper no matter the results (as an in-principle acceptance, thus if there are major unjustified deviations from what was planned, the paper may be rejected). The results from Scheel et al. (2021), who compare Registered Reports to normal publications, suggest that the former publication track leads to more null results than the latter. Whether this is necessarily the case in the long run is not clear—the Registered Report format puts the focus on the hypotheses and designs, which can potentially lead to more interesting hypotheses being tested with better designs, and whether this will lead to null results is not obvious but it does help against publication bias. Registered Reports are getting more common in many fields, including in economics where there are now a handful of journals offering such tracks.

9 Discussion

So far we have focused on reproducibility or replicability. What about generalizability? Whether your results of the general hypothesis you are testing will replicate if you change the population, design, or analytical choices, all relate to generalizability and potential heterogeneity. To what extent results from student samples generalize or are externally valid has perhaps received the most attention thus far, with studies such as Henrich et al. (2004) exploring economic games in small-scale societies and Snowberg and Yariv (2021) comparing student samples to a US representative sample and an online sample. Depending on the setting and game, there can be substantial heterogeneity.

As has hopefully come across, we believe there is real upside to pre-analysis plans. We may disagree on how valuable they are currently—Anna is very positive, and Lucas is more skeptical about how they are used (both by referees and researchers)—but we hope they can be a part of improving our joint production of knowledge. When it comes to the generalizability of the results from a study with a pre-analysis plan, there are still limits to consider. Think about all the potential ways to test a hypothesis for a given dataset as the garden of forking paths (as in Gelman and Loken, 2013). A pre-analysis plan shows one potential fork in the garden of forking paths, and since this is pre-registered, it also gives rise to meaningful p-values. However, there are typically many forks possible, and different researchers might choose different forks when asked to write a pre-analysis plan for the same hypothesis and data. Results from such pre-analysis plans should not result in any systematic bias in effect sizes, but if there is variation in chosen forks they will underestimate the standard error in the statistical test as the standard error does not incorporate this additional heterogeneity. A number of recent "multi-analyst" studies where many researchers are asked to test the same hypotheses on the same data suggest that there can be substantial variation in analysis paths and results (see, e.g., Silberzahn et al., 2018; Botvinik-Nezer et al., 2020; Menkveld et al., 2024)—depending on who you ask you can sometime get statistically significant results in different

directions with the same data. This analytical heterogeneity should however be smaller for experimental data than for many other types of empirical data in economics.

A study testing a given hypothesis can typically also be designed in different ways that can all be considered appropriate to test the hypothesis. To what extent there is heterogeneity in designs is to some extent related to conceptual replications. A paper exploring this topic is Huber et al. (2023) who asked different research teams to design studies to test the same research question "Does competition affect moral behavior?", keeping the analysis constant. The results suggest substantial design heterogeneity, which is 1.6 times as large as the average standard error of effect size estimates. See Holzmeister et al. (2024) for more discussion on population heterogeneity, analytical heterogeneity and design heterogeneity. A clear takeaway from this literature is that generalizability based on single designs can be limited, and maybe we should reward authors more for clearly discussing the limitations of their studies. It is interesting to note that in journals in other fields such as psychology, authors are asked to include a section on limitations. When writing your paper, if you believe that the results are likely to hold only for e.g. gender equal or gender unequal countries, it is not only helpful for the reader, it is in the interest of the researcher.

Finally, you're reading this chapter likely as a new experimentalist hoping to do good science. Though we have provided a lot of details, and maybe too much editorializing, we do want to end where we began. We hope you'll be rigorous in producing true and robust results. In other words, "Make sure it's right the first time." And help others understand your work well enough that they can replicate and reproduce everything in a manner you'd appreciate. So, "be very clear and very comprehensive in describing everything you did." Following these simple rules, you'll be confident in and proud of your own work, and you'll be confident when others attempt to replicate your impactful results.

10 References for pre-analysis plans

10.1 Templates

Many journals have templates. You should check the journal you are planning to submit to. Here is the template from JPE: Micro, a journal that requires a pre-analysis plan: https://www.journals.uchicago.edu/journals/jpemi/data-generation-guidelines

Open Science Framework:
https://help.osf.io/article/229-select-a-registration-template

10.2 Outlets where you can post

American Economic Association Social Science Registry:
https://www.socialscienceregistry.org/

As Predicted: https://aspredicted.org

Open Science Framework:
https://help.osf.io/article/158-create-a-preregistration

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